野火_ZYNQ7010 BTB 底板_原理图_V1.1

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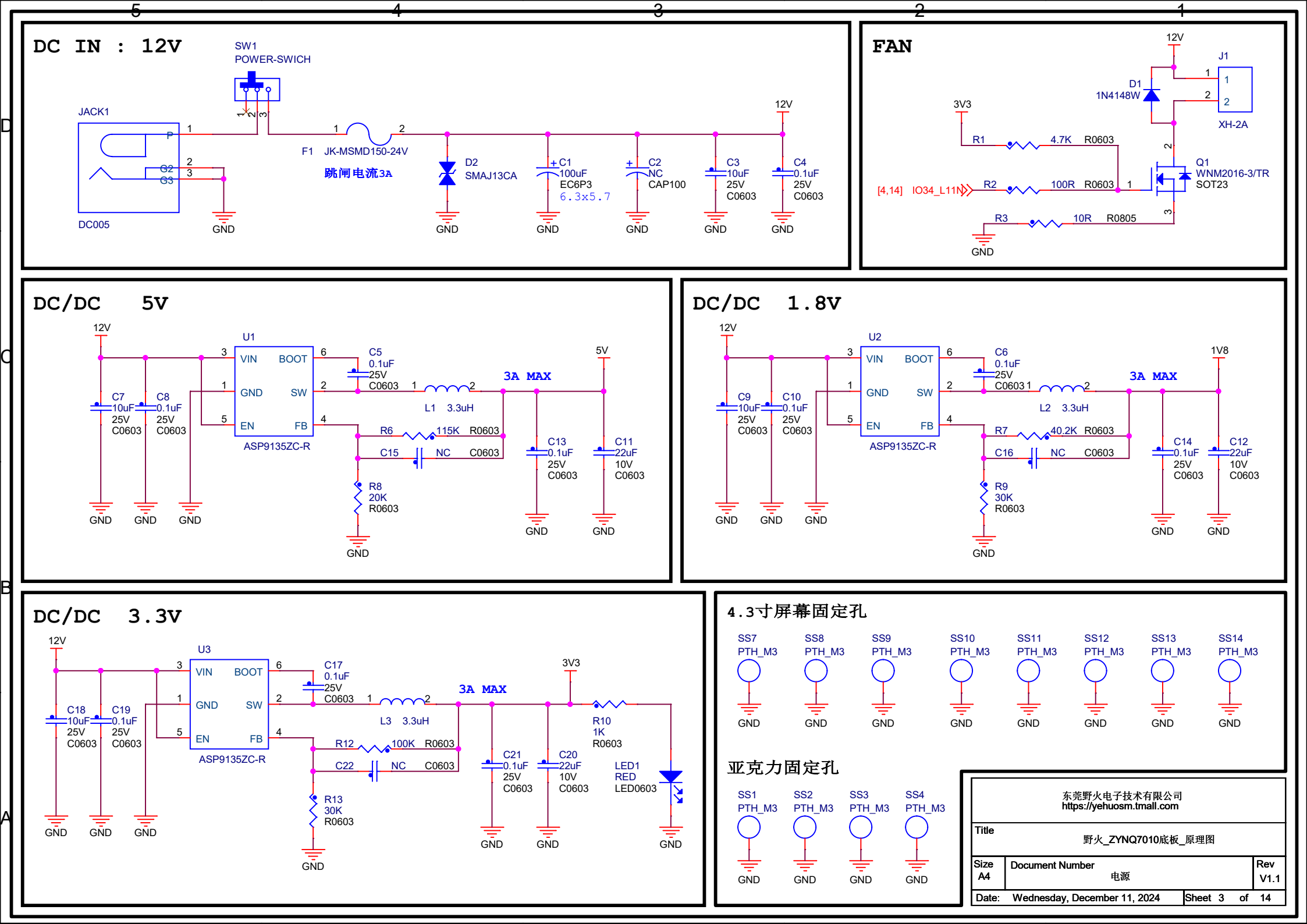
Title: 野火_ZYNQ7010底板_原理图

Size: A4 | Document Number: 目录 | Rev: V1.1

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历史版本

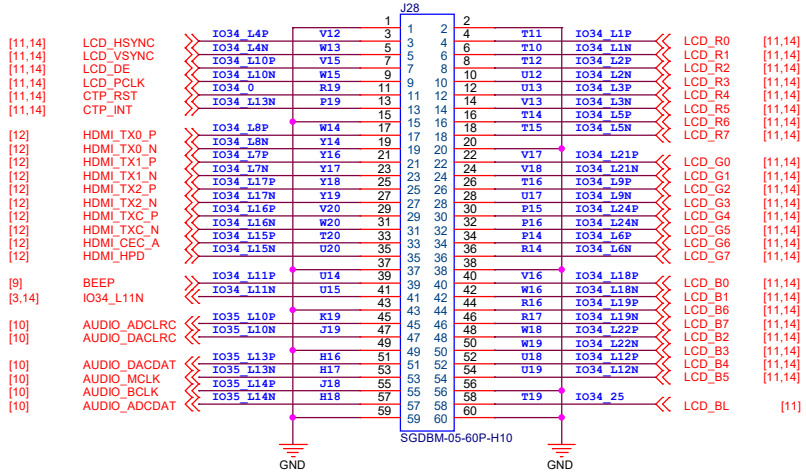
版本号	日期	设计	描述
V0.1	2020-10-26	xgh	初始版本
V1.0	2023-06-26	ghs	BTB初始版本
V1.1	2023-09-19	ghs	修正JTAG接口定义错误，修正音频接口封装错误

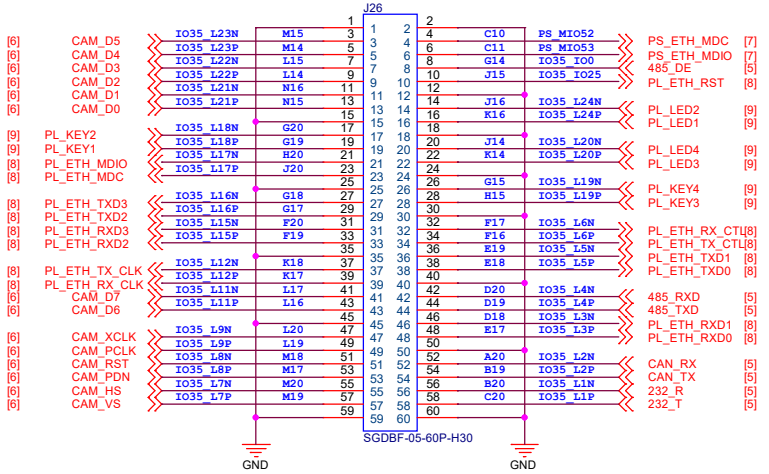

DC IN : 12V
SW1
POWER-SWICH
JACK1
DC005
F1 JK-MSMD150-24V
跳闸电流3A
D2
SMAJ13CA
C1 100uF EC6P3 6.3x5.7
C2 NC CAP100
C3 10uF 25V C0603
C4 0.1uF 25V C0603
GND
12V
FAN
J1
XH-2A
D1
1N4148W
3V3
R1 4.7K R0603
R2 100R R0603
R3 10R R0805
[4,14] IO34_L11N
Q1
WNM2016-3/TR
SOT23
DC/DC 5V
U1
ASP9135ZC-R
VIN
BOOT
GND
SW
EN
FB
C5 0.1uF 25V C0603
L1 3.3uH
3A MAX
5V
R6 115K R0603
C15 NC C0603
R8 20K R0603
C7 10uF 25V C0603
C8 0.1uF 25V C0603
C13 0.1uF 25V C0603
C11 22uF 10V C0603
DC/DC 1.8V
U2
C6 0.1uF 25V C0603
L2 3.3uH
1V8
R7 40.2K R0603
C16 NC C0603
R9 30K R0603
C9 10uF 25V C0603
C10 0.1uF 25V C0603
C14 0.1uF 25V C0603
C12 22uF 10V C0603
DC/DC 3.3V
U3
C17 0.1uF 25V C0603
L3 3.3uH
R12 100K R0603
C22 NC C0603
R13 30K R0603
R10 1K R0603
C18 10uF 25V C0603
C19 0.1uF 25V C0603
C21 0.1uF 25V C0603
C20 22uF 10V C0603
LED1 RED LED0603
4.3寸屏幕固定孔
SS7 PTH_M3
SS8 PTH_M3
SS9 PTH_M3
SS10 PTH_M3
SS11 PTH_M3
SS12 PTH_M3
SS13 PTH_M3
SS14 PTH_M3
亚克力固定孔
SS1 PTH_M3
SS2 PTH_M3
SS3 PTH_M3
SS4 PTH_M3
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野火_ZYNQ7010底板_原理图
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电源
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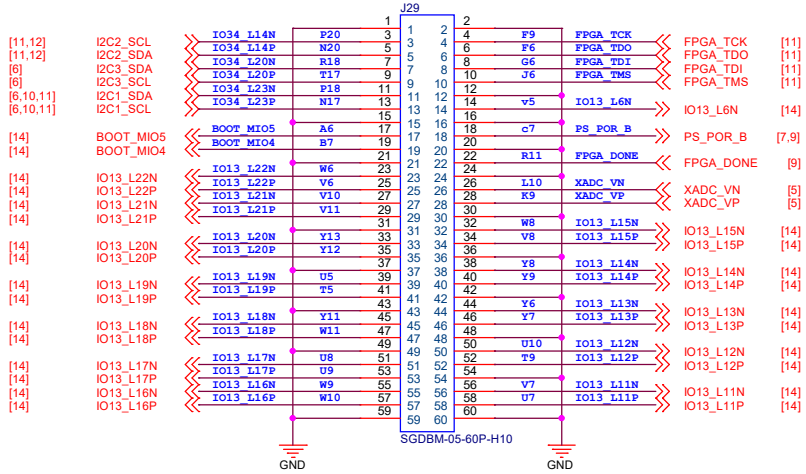
BTB Connector

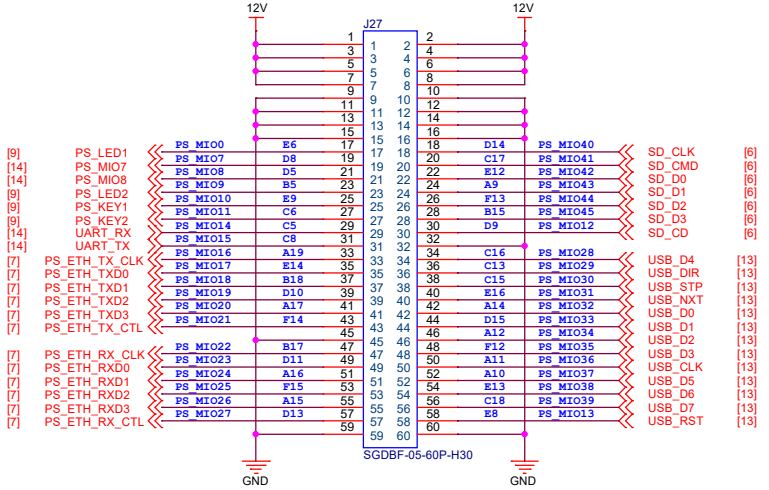
60*4个引脚包含

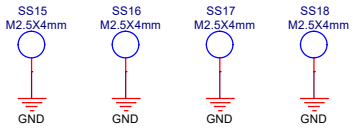
XADC	2
POR	1
DONE	1
PS_IO	42
PL_IO	124
JTAG	4
12V输入	4
GND	55
BOOT	2





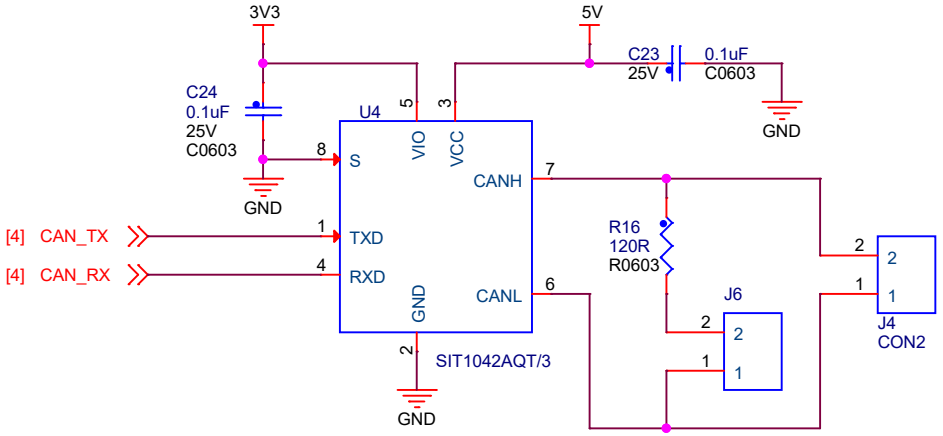




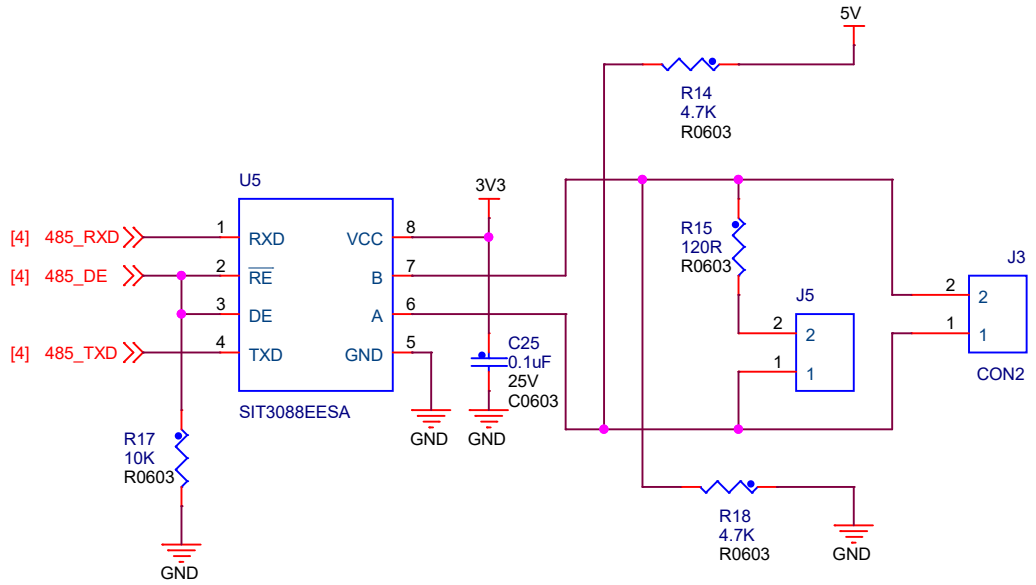


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Title: 野火_ZYNQ7010底板_原理图		
Size: A3	Document Number: BTB Connector	Rev: V1.1
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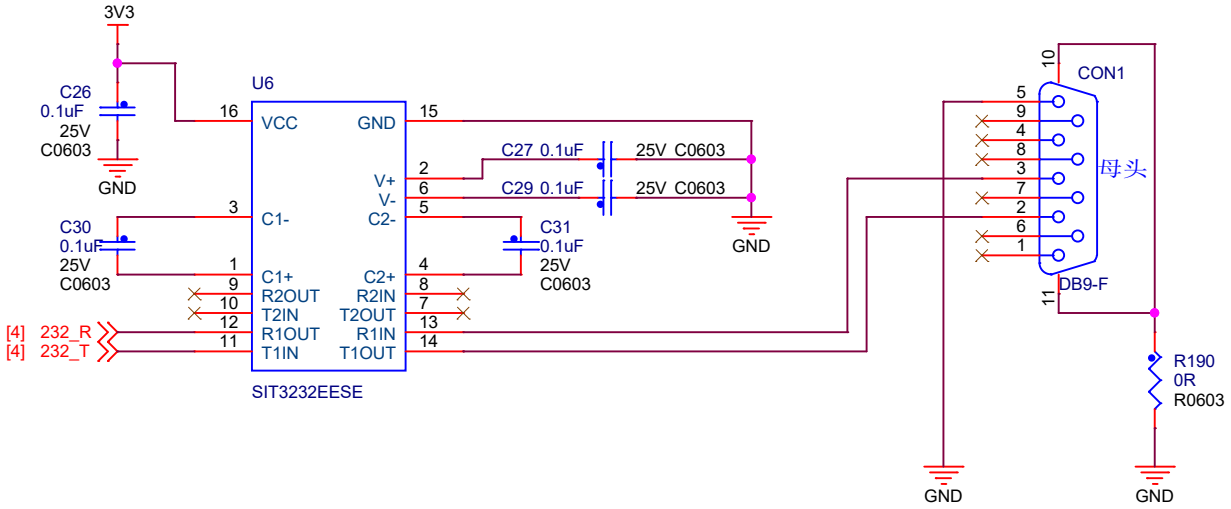
CAN



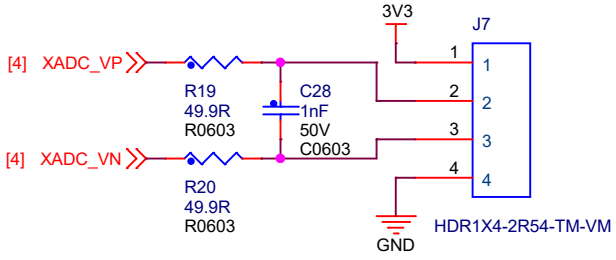
RS485



RS232串口

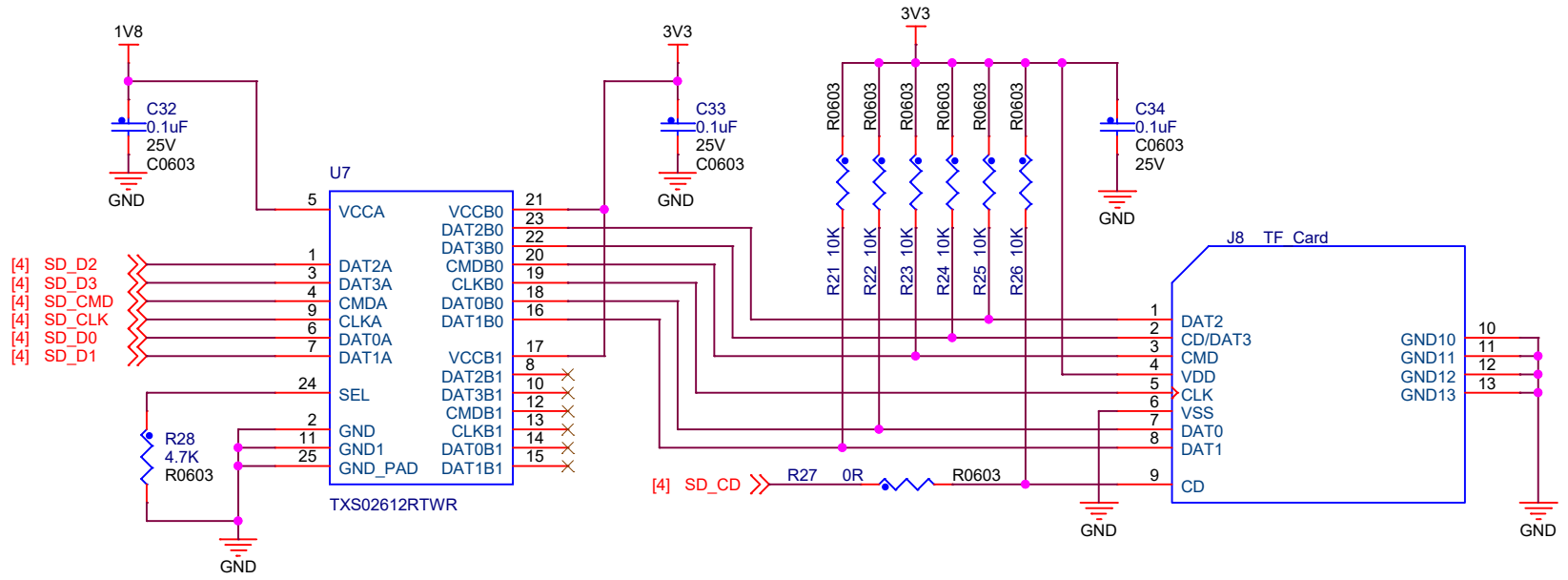


ADC



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Title	野火_ZYNQ7010底板_原理图	
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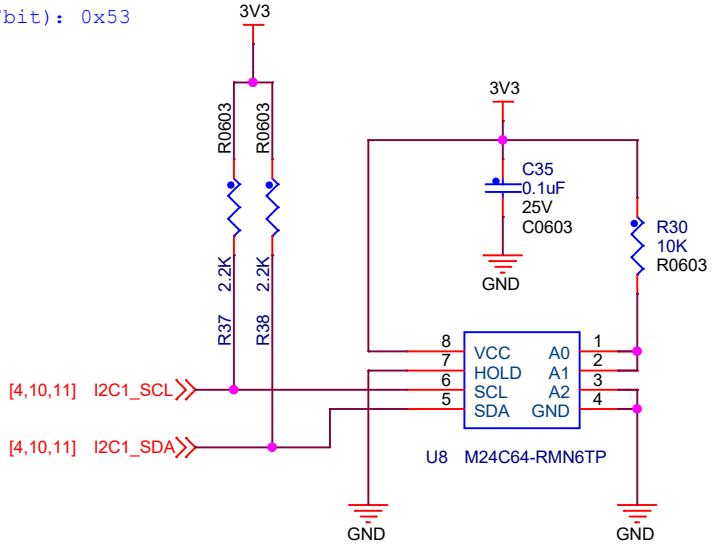
SD卡



EEPROM

容量：8K字节

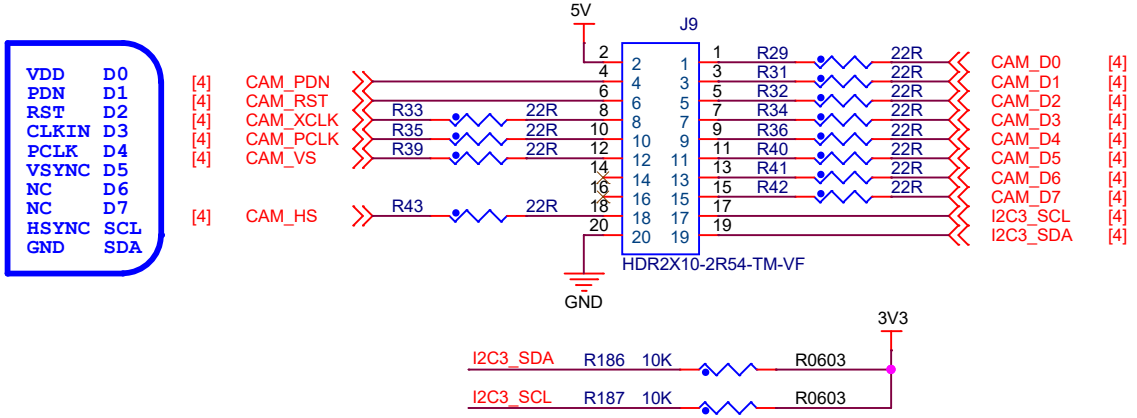
I2C地址(7bit)：0x53



摄像头接口

OV5640_I2C地址(7bit)：0x3C

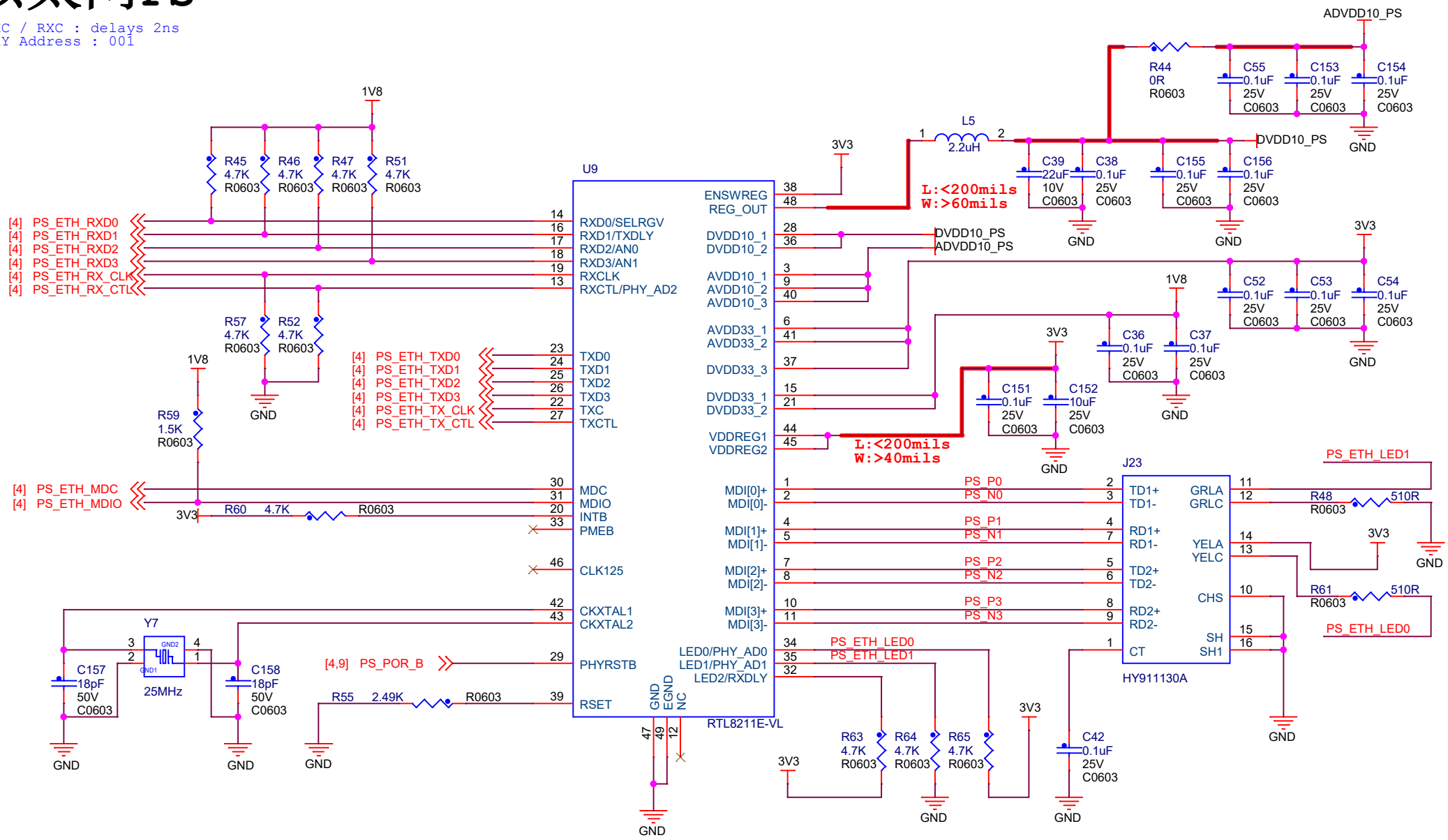
OV7725_I2C地址(7bit)：0x21



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以太网PS

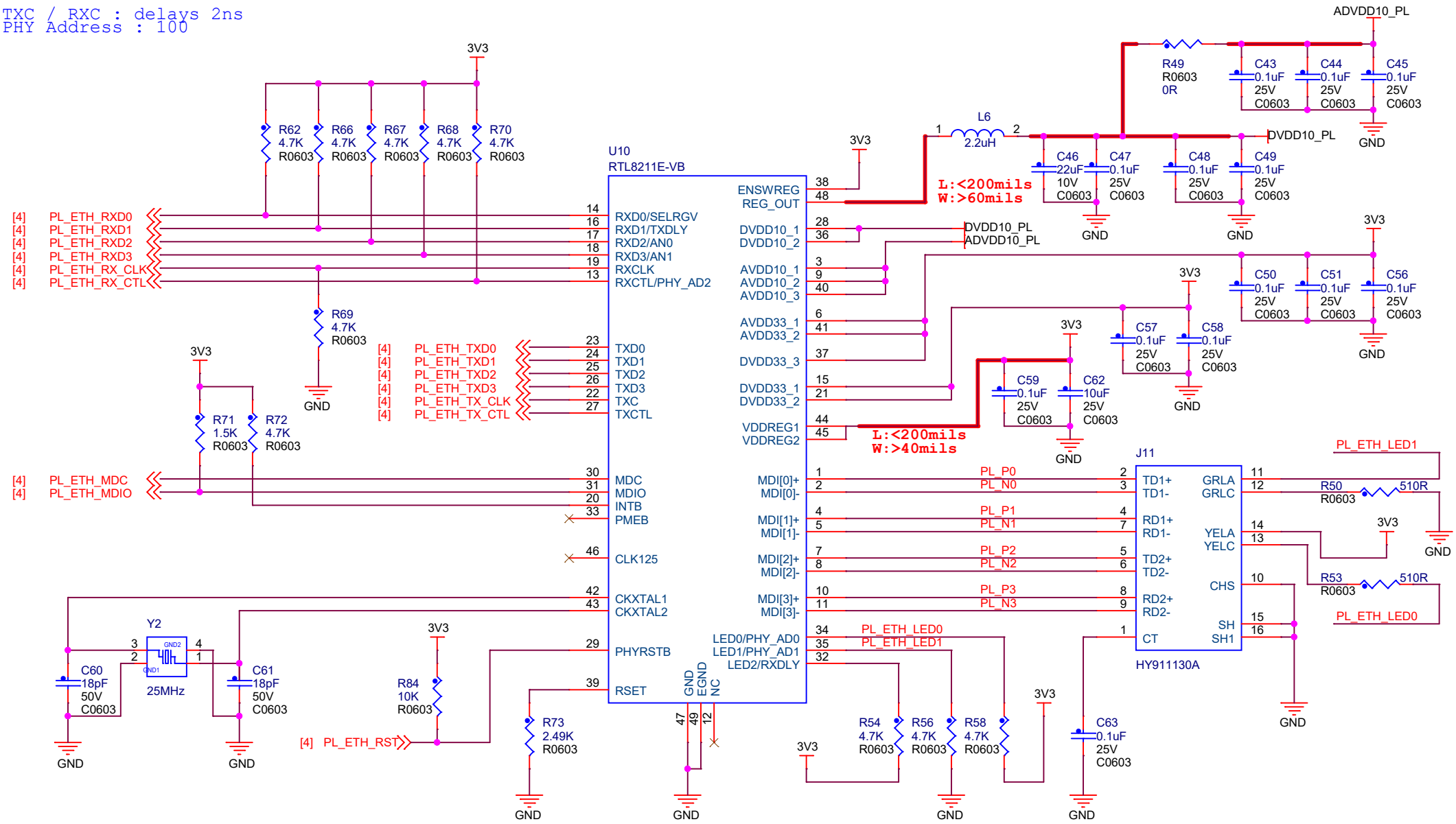
TXC / RXC : delays 2ns
PHY Address : 001



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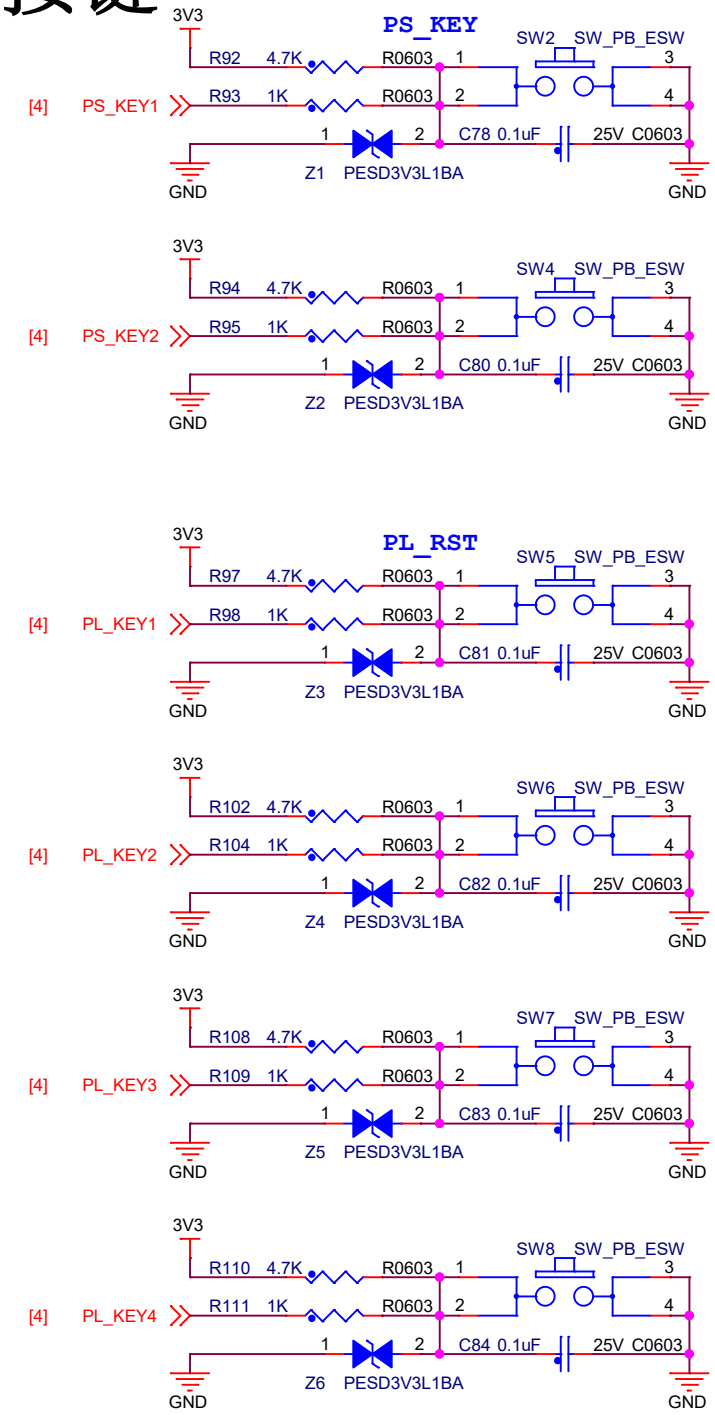
以太网接口PL

TXC / RXC : delays 2ns
PHY Address : 100

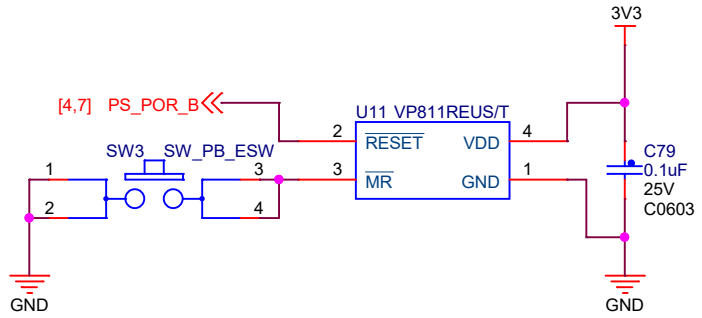


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Title	野火_ZYNQ7010底板_原理图
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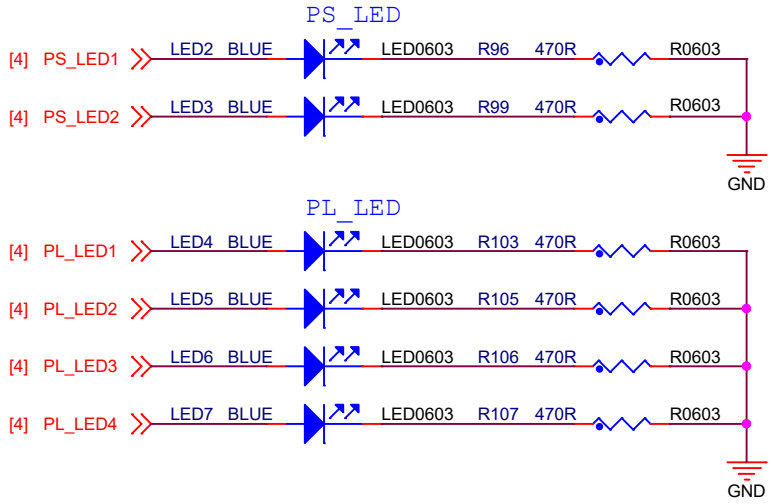
按键



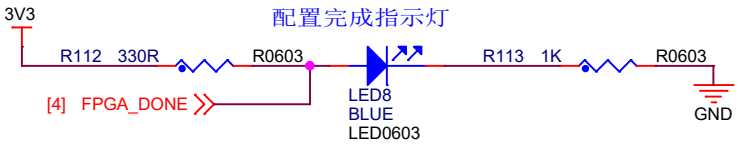
重配置按键



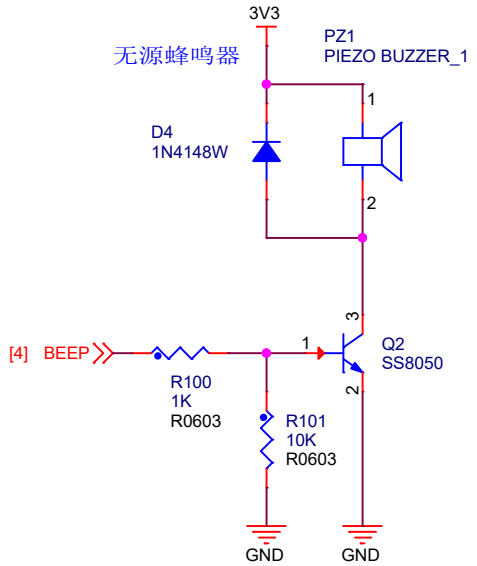
USER LED



Configure LED

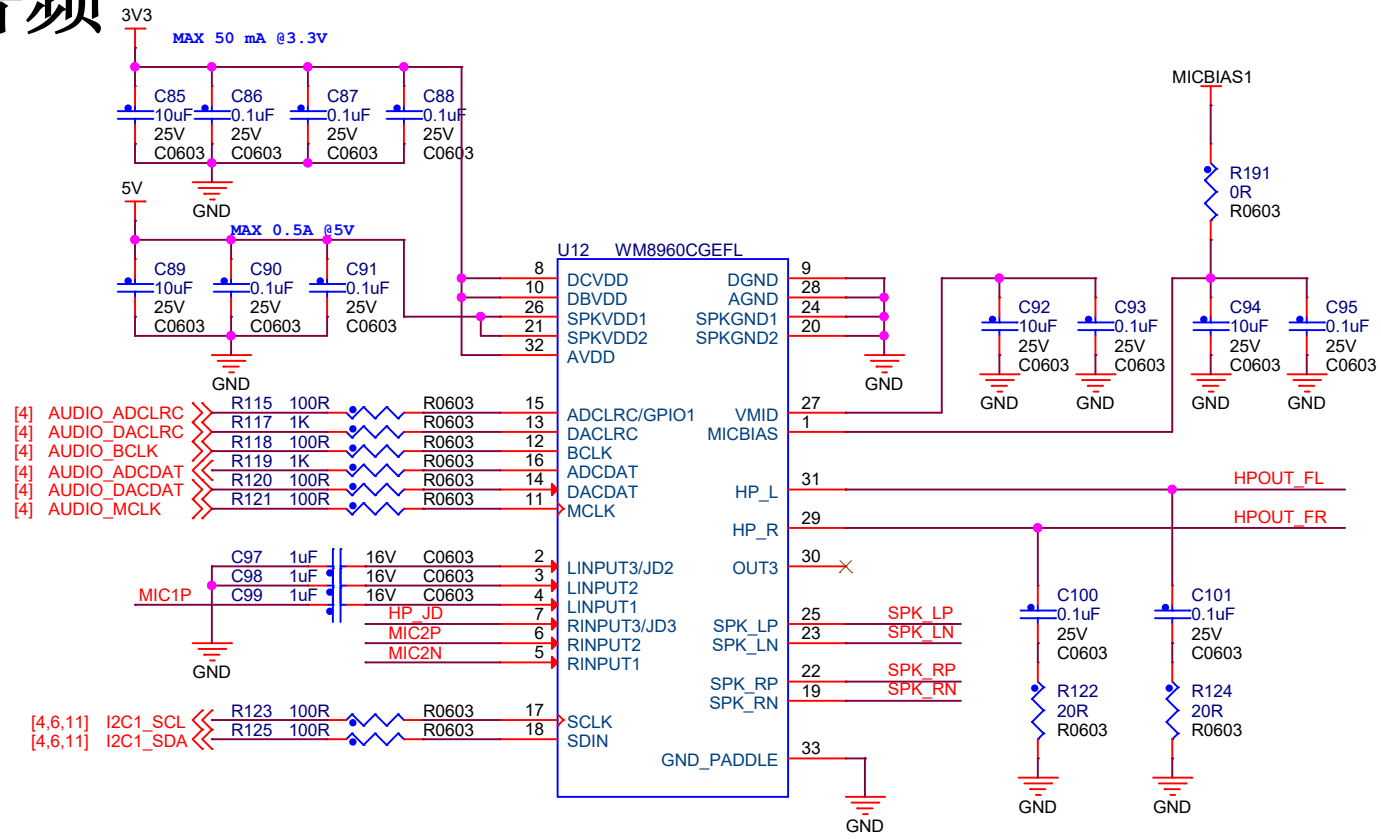


蜂鸣器

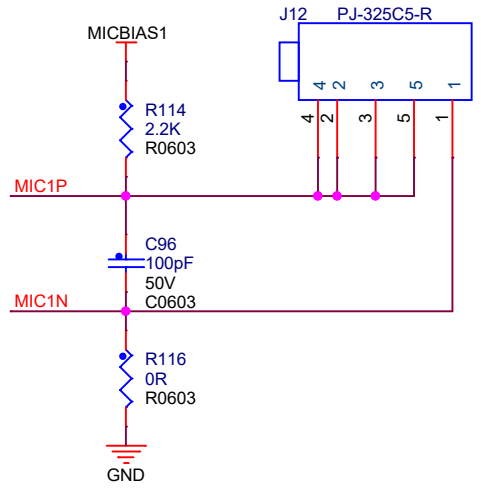


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Size A4	Document Number 蜂鸣器器/按键/LED
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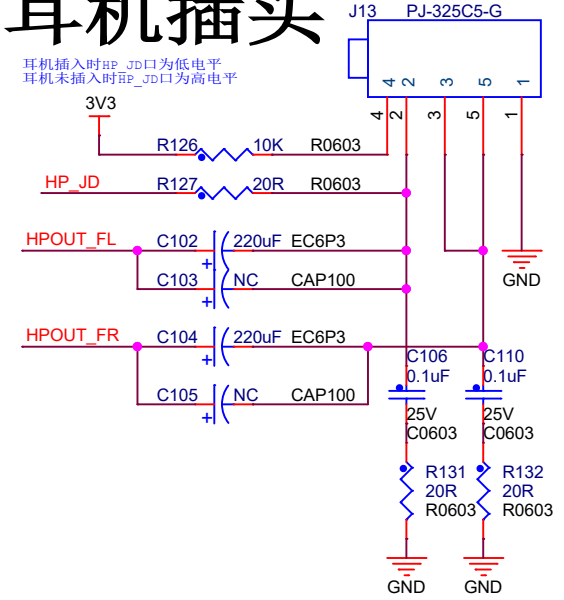
音频



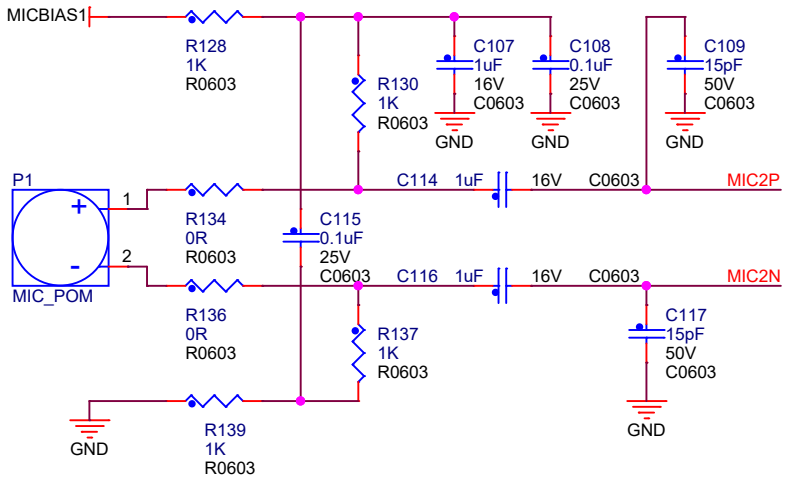
MIC插头



耳机插头

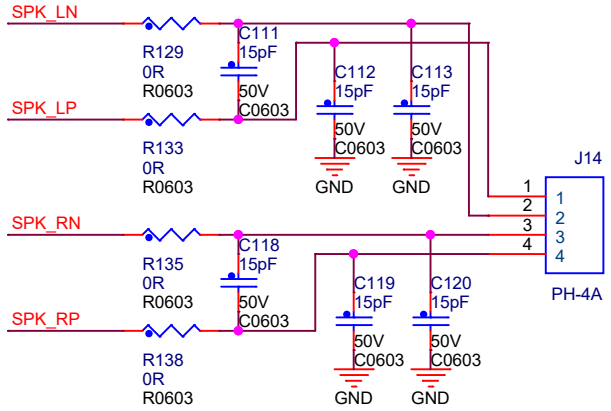


MIC



喇叭插座

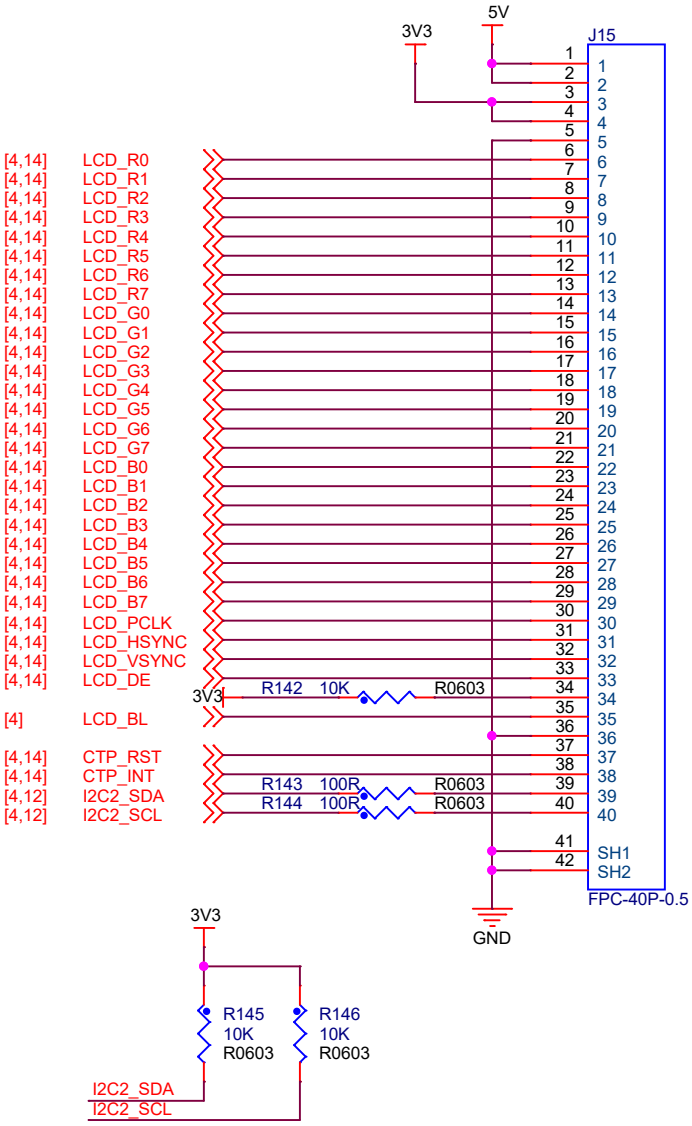
可以驱动2个8Ω 1W的喇叭（BTL方式）



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Title	野火_ZYNQ7010底板_原理图	
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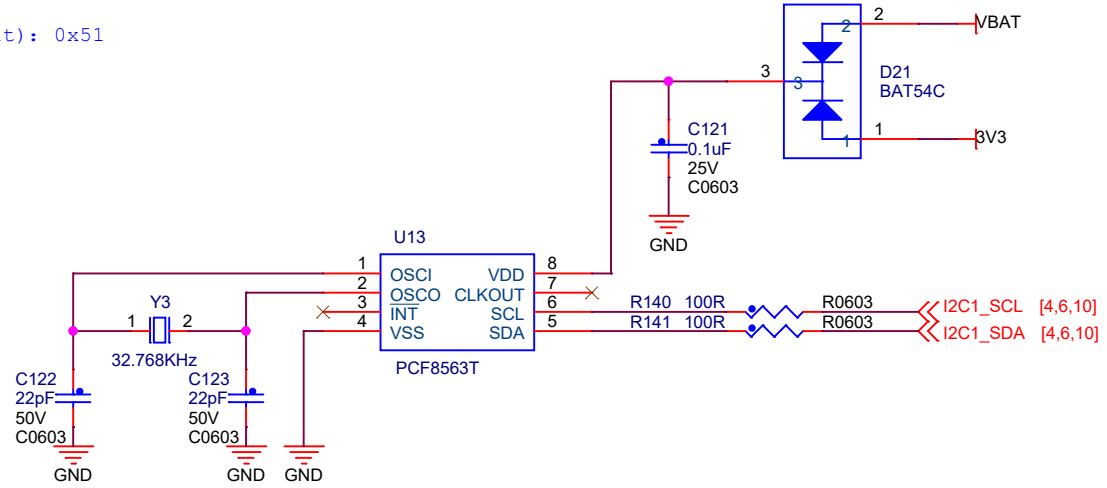
LCD RGB888

I2C地址(7bit)：0x5D

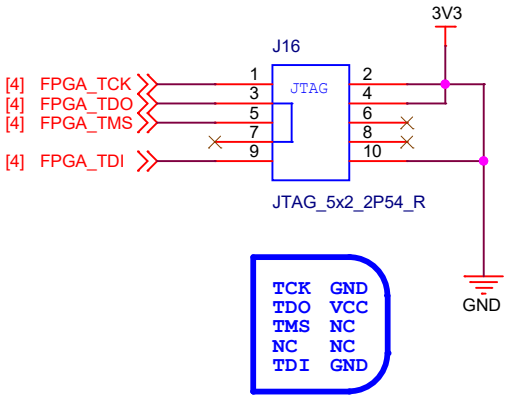


RTC

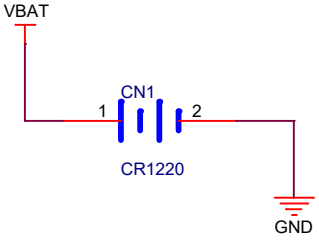
I2C地址(7bit)：0x51



JTAG调试口



RTC电池

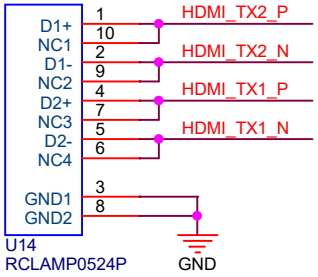


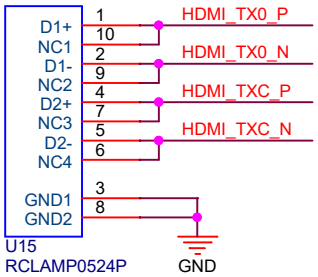
东莞野火电子技术有限公司 https://yehuosm.tmall.com	
Title	野火_ZYNQ7010底板_原理图
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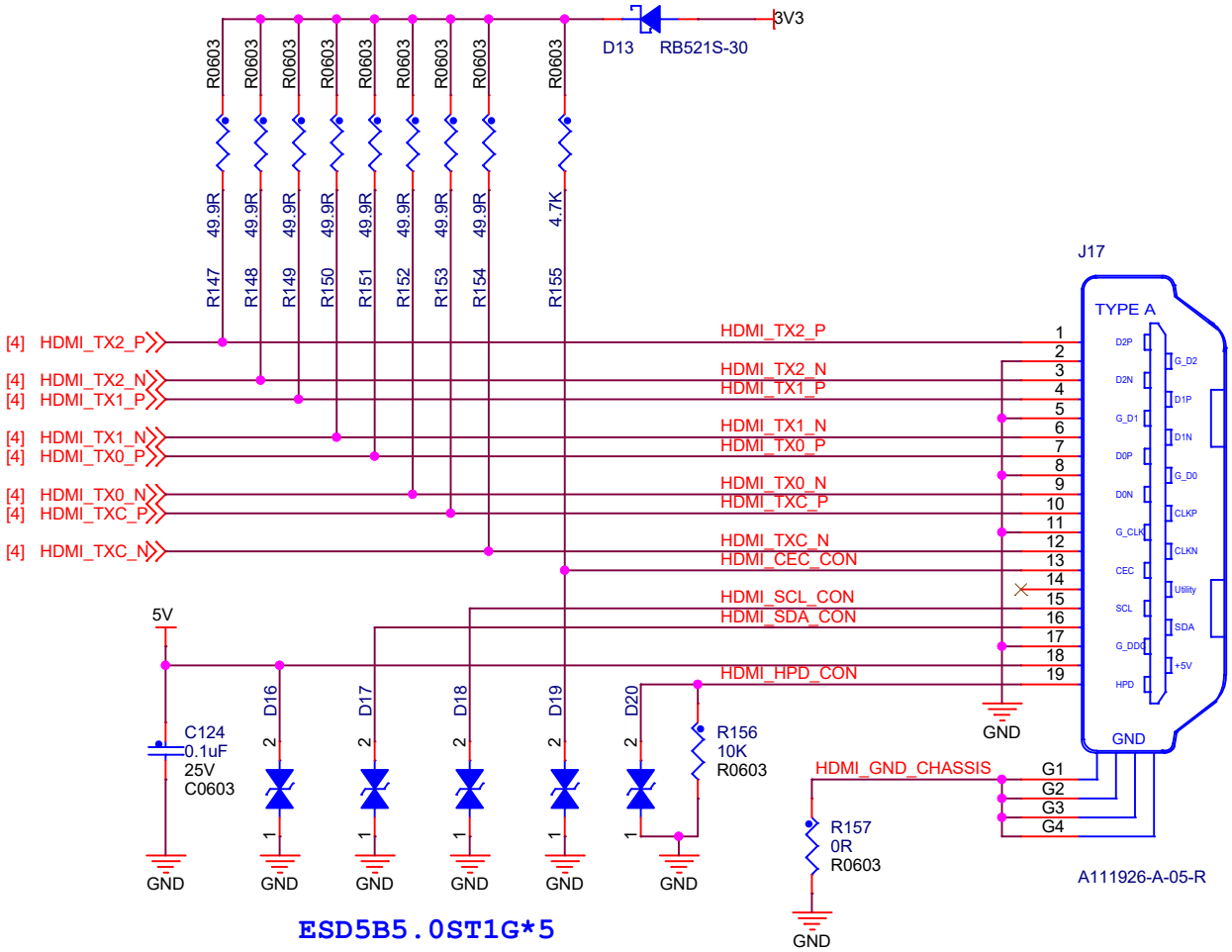
HDMI连接器

Type A型

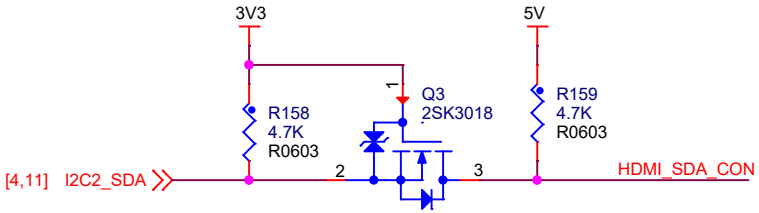
I2C地址(7bit)：0x50

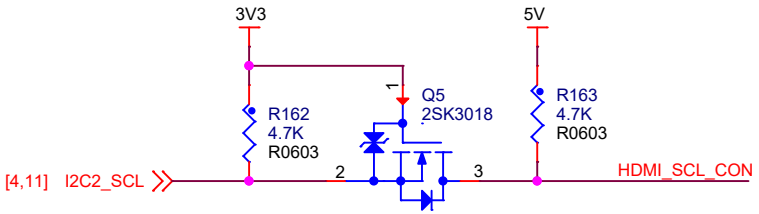


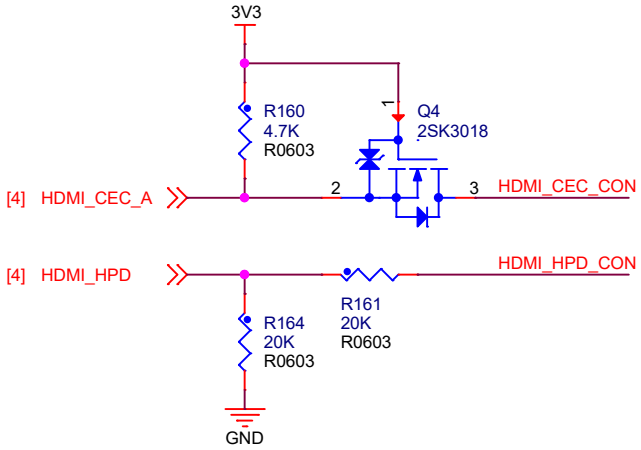




ESD5B5.0ST1G*5

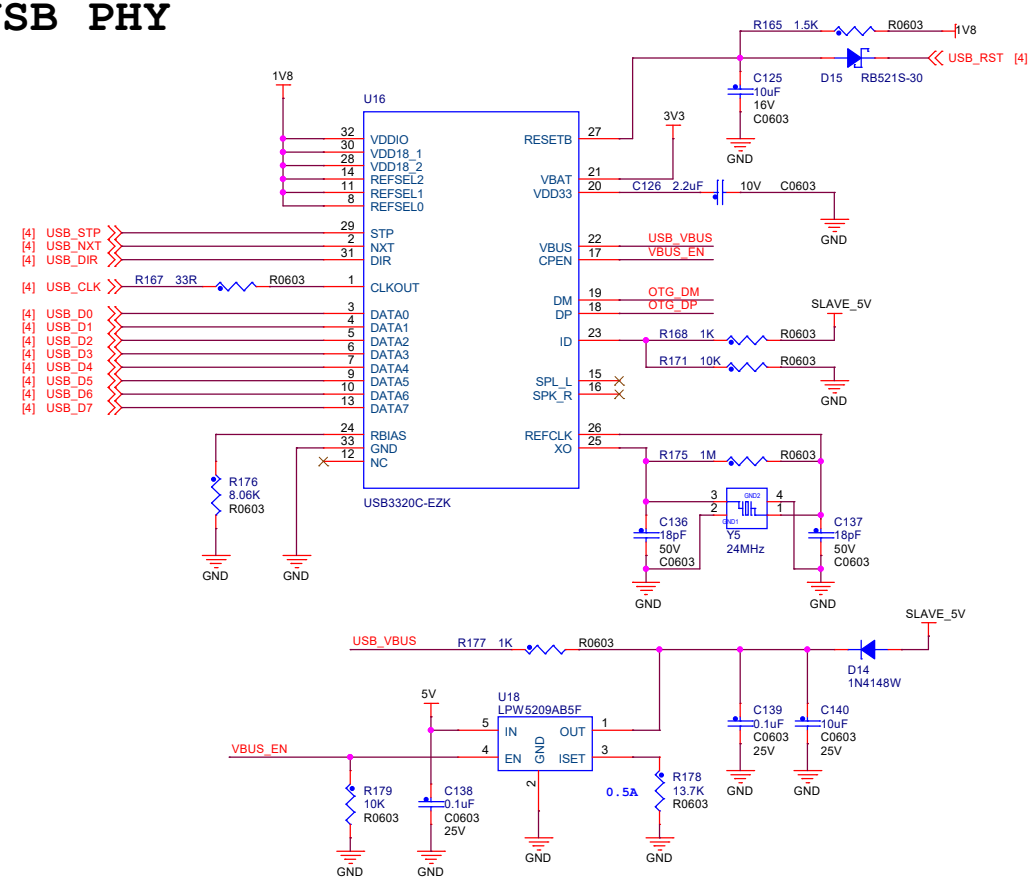




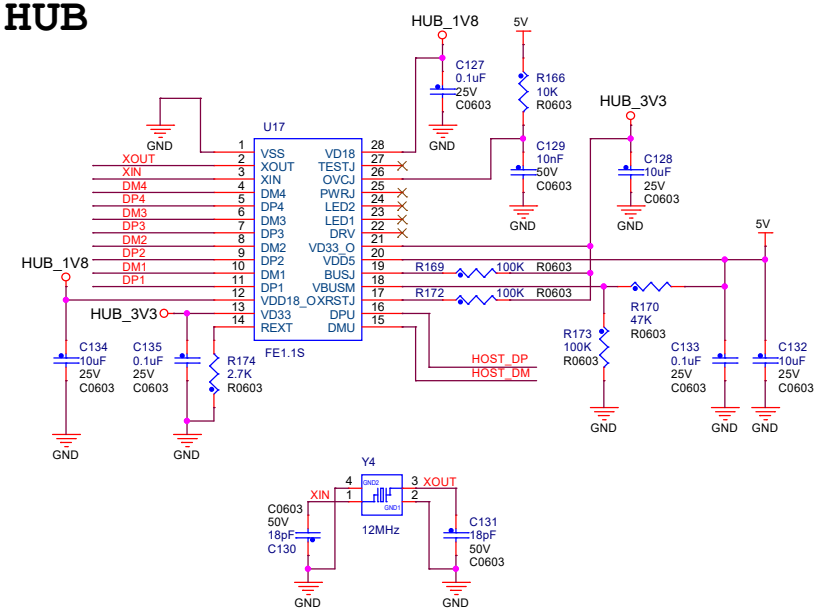


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Title	野火_ZYNQ7010底板_原理图	
Size A4	Document Number HDMI PORT	Rev V1.1
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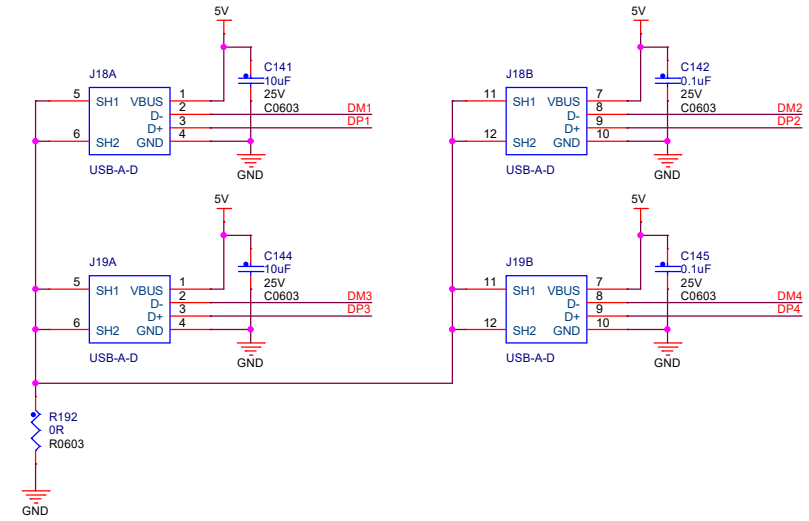
USB PHY



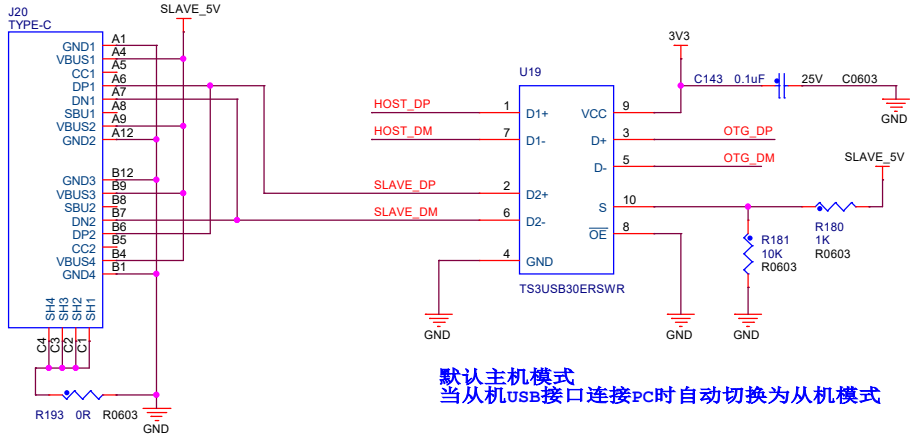
USB HUB



USB HOST

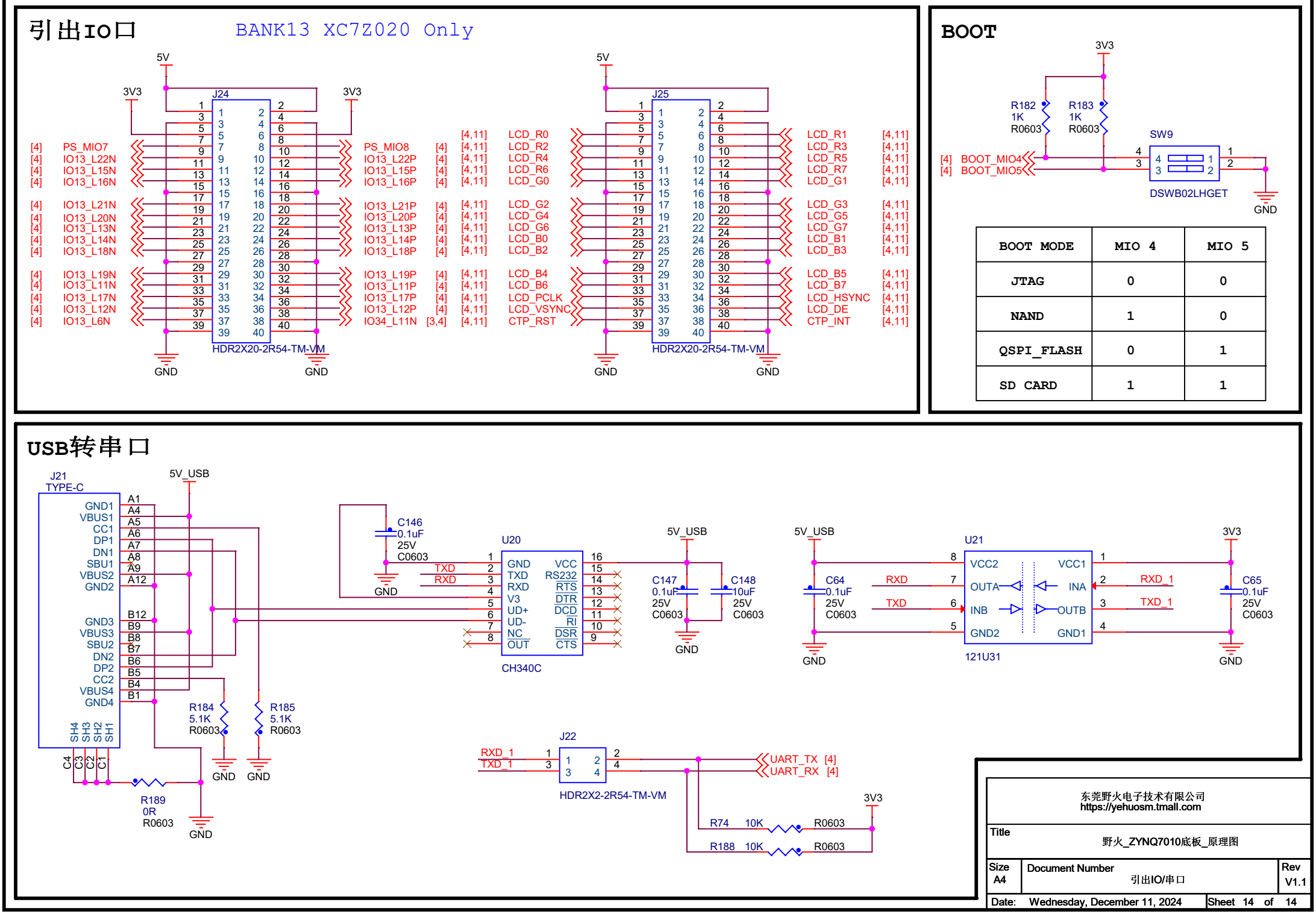


USB SLAVE



默认主机模式
当从机USB接口连接PC时自动切换为从机模式

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Size A3	Document Number USB PHY/HUB	Rev V1.1
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引出IO口
BANK13 XC7Z020 Only
J24
J25
HDR2X20-2R54-TM-VM
BOOT
SW9
DSWB02LHGET
BOOT MODE | MIO 4 | MIO 5
JTAG | 0 | 0
NAND | 1 | 0
QSPI_FLASH | 0 | 1
SD CARD | 1 | 1
USB转串口
J21
TYPE-C
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